Observed Monitoring Metrics — ML-Based Traffic Speed Prediction

Companion document to the paper "Cloud-Fog-Edge Deployment of ML-Based Traffic Speed Prediction: An Empirical Analysis of Performance, Cost, and Scalability". It reports the resource-utilisation, thermal, and power metrics observed during the experiments, which the paper references but does not present in full.

Collection methodology

A background monitoring process sampled system metrics every 0.5 seconds during each experiment. CPU variables: system-wide utilisation (%), per-process utilisation (%), operating frequency (MHz), package temperature (deg C), and — where accessible — power (W) via Intel RAPL. GPU variables (via nvidia-smi, on GPU-equipped platforms only): utilisation (%), memory use (MB), temperature (deg C), and power (W).

Reliability caveats (important)

1. **Attribution.** CPU-system and GPU readings are whole-node/whole-device: on shared or virtualised platforms (CSF3 nodes, AWS EC2) they may include activity not attributable to the experiment process. They should be read as qualitative context, not precise per-process measurements; repetition of runs does not remove this systematic attribution error. Only the Dell Precision platform (dedicated hardware, no virtualisation) provides fully representative CPU and GPU readings.
2. **Power.** CPU power via Intel RAPL was accessible only on the Dell Precision (average CPU power 10-46 W across tiers; 0.1-8.7 Wh per training run). On CSF3 the powercap energy counters are root-restricted; AWS EC2 does not expose hardware power to guests. GPU power (nvidia-smi) is available on the GPU platforms.
3. **Diagnostic value.** The telemetry proved essential for outlier attribution: inflated fog-platform runs in an early campaign showed mean CPU clocks of 0.5-1.6 GHz at low package temperature — the signature of an OS power-saving state, not thermal throttling or contention — and were re-measured on mains power. On AWS t2.medium, elevated steal time and run-position-dependent slowdowns identified burstable CPU-credit exhaustion.

Observed metrics by platform and sensor tier

Values aggregate the repeated runs of each platform's initial-training batch: mean of per-run averages; "max" is the maximum across runs. Empty cells: metric not available on that platform.

CSF3 Cloud GPU (A100)

Tier	CPU sys avg %	CPU proc avg %	CPU freq avg MHz	CPU temp avg/max C	GPU util avg %	GPU mem avg MB	GPU power avg/max W	GPU temp avg/max C
tiny	19.4	147.2	2,892	45.0/46.8	6.1	16,730	65.2/83.2	25.0/26.0
small	16.7	134.7	2,889	44.7/48.0	7.5	16,834	65.3/67.8	25.6/26.0
medium	16.6	130.1	2,887	43.7/47.2	7.5	16,861	65.0/81.9	24.8/26.0
full	15.7	125.2	2,873	42.6/45.8	7.2	16,907	64.9/68.1	24.5/25.0

Dell Precision Fog GPU (RTX 4090)

Tier	CPU sys avg %	CPU proc avg %	CPU freq avg MHz	CPU temp avg/max C	GPU util avg %	GPU mem avg MB	GPU power avg/max W	GPU temp avg/max C
tiny	13.1	180.0	1,967	83.3/94.0	5.4	14,373	28.7/31.9	56.1/58.0
small	10.9	150.2	2,189	86.3/100.0	7.4	14,450	30.3/32.5	59.2/60.0
medium	11.1	152.0	2,163	86.6/100.0	6.9	14,477	30.7/33.4	59.7/62.0
full	10.0	134.1	2,127	86.3/100.0	6.7	14,607	30.4/34.0	59.6/61.0

CSF3 Jetson Sim

Tier	CPU sys avg %	CPU proc avg %	CPU freq avg MHz	CPU temp avg/max C	GPU util avg %	GPU mem avg MB	GPU power avg/max W	GPU temp avg/max C
tiny	9.5	127.2	2,785	47.7/49.5	4.8	8,596	68.1/70.4	25.9/26.0
small	9.3	123.5	2,801	49.4/51.0	6.3	8,643	68.7/81.1	26.5/27.0

CSF3 Edge RSU Sim

Tier	CPU sys avg %	CPU proc avg %	CPU freq avg MHz	CPU temp avg/max C	GPU util avg %	GPU mem avg MB	GPU power avg/max W	GPU temp avg/max C
tiny	62.4	93.1	2,250	84.2/85.0	—	—	—/—	—/—
small	60.5	90.8	2,250	84.1/86.2	—	—	—/—	—/—

CSF3 RPi Sim

Tier	CPU sys avg %	CPU proc avg %	CPU freq avg MHz	CPU temp avg/max C	GPU util avg %	GPU mem avg MB	GPU power avg/max W	GPU temp avg/max C
tiny	85.9	131.5	2,250	84.7/86.0	—	—	—/—	—/—
small	78.0	138.2	2,250	84.3/86.6	—	—	—/—	—/—

AWS t2.medium RSU

Tier	CPU sys avg %	CPU proc avg %	CPU freq avg MHz	CPU temp avg/max C	GPU util avg %	GPU mem avg MB	GPU power avg/max W	GPU temp avg/max C
tiny	64.9	128.9	2,300	—/—	—	—	—/—	—/—
small	63.6	125.9	2,300	—/—	—	—	—/—	—/—

Interpretation notes

- CPU process utilisation above 100% indicates multi-threaded data loading and feature engineering phases; training itself is GPU-bound on GPU platforms (GPU utilisation averages are low because per-step kernels for a ~142k-parameter MLP are dominated by launch overhead, consistent with the paper's finding that tensor-core throughput is not the bottleneck).
- On the fog laptop workstation, sustained combined CPU+GPU load settles at reduced CPU clocks (1.2-1.7 GHz) at 80-88 deg C: normal power sharing that does not measurably affect GPU-bound wall-clock times.
- The AWS t2.medium rows reflect a burstable instance: sustained full-CPU work beyond ~1.5 hours exhausts CPU credits and caps the instance at its baseline rate; short workloads run at burst speed. Deployment schedules on burstable instances should be budgeted at baseline speed.